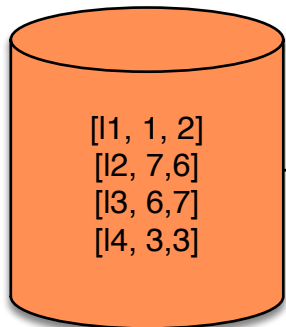


### Premier fragment matrice



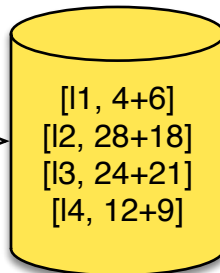
$V_1[4,3]$

read()

Mapper 1

emit()

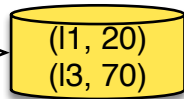
### Fragments intermédiaires



(1, 10)  
(3, 45)

Reducer 1

(1, 10+10)  
(3, 45+25)



### Regroupement par ligne

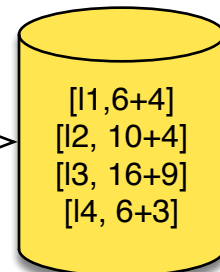
### Résultat final

$V_2[2,1]$

read()

Mapper 2

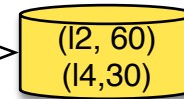
emit()



(2, 14)  
(4, 9)

Reducer 2

(2, 46+14)  
(4, 21+9)



### Distribution par ligne

### Second fragment matrice

